FABRIC MANUFACTURING

Part: B

Course Code: FE 253-0723

Weft Knitting Machines

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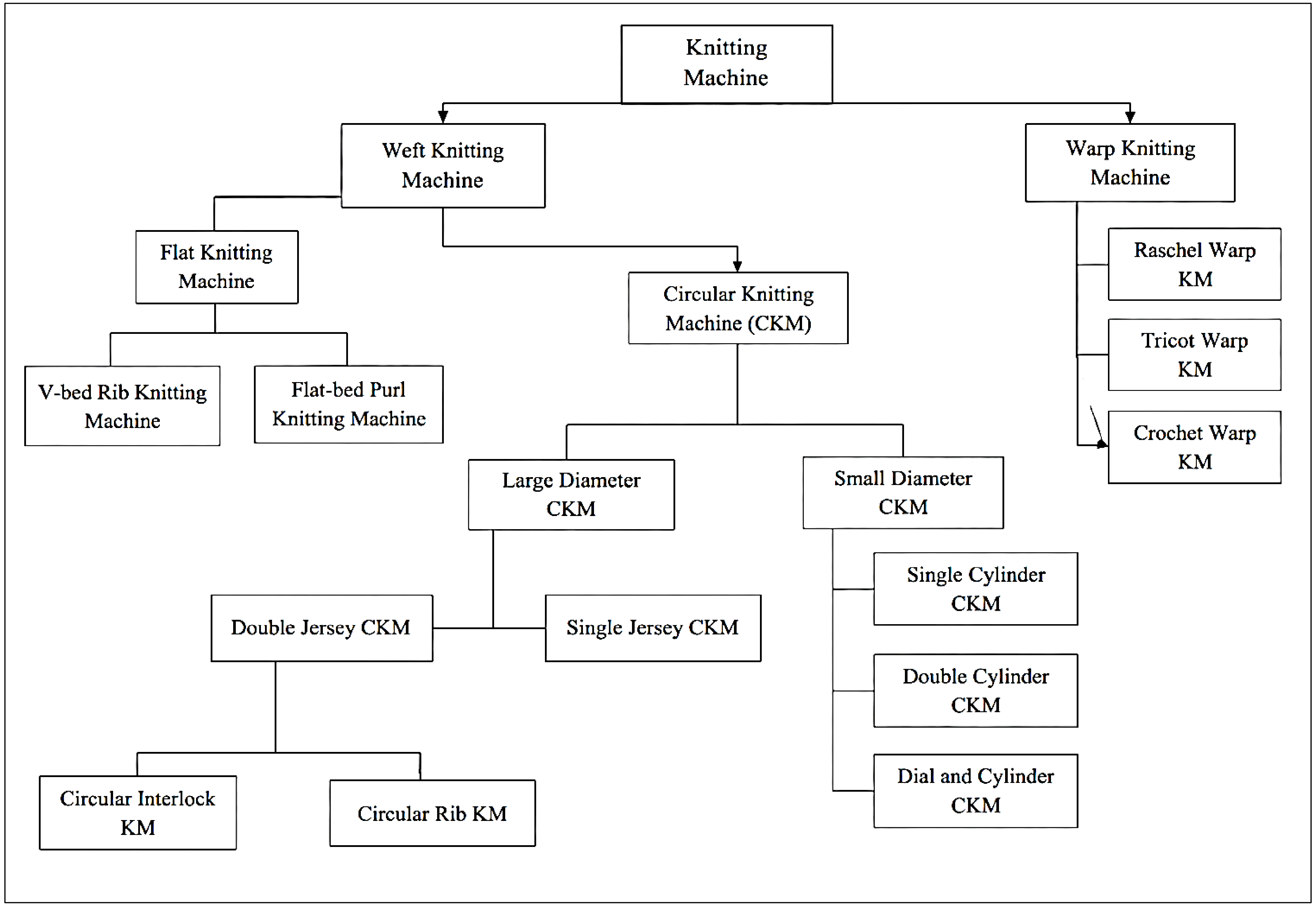
CLASSIFICATIONS



TYPES OF MACHINE

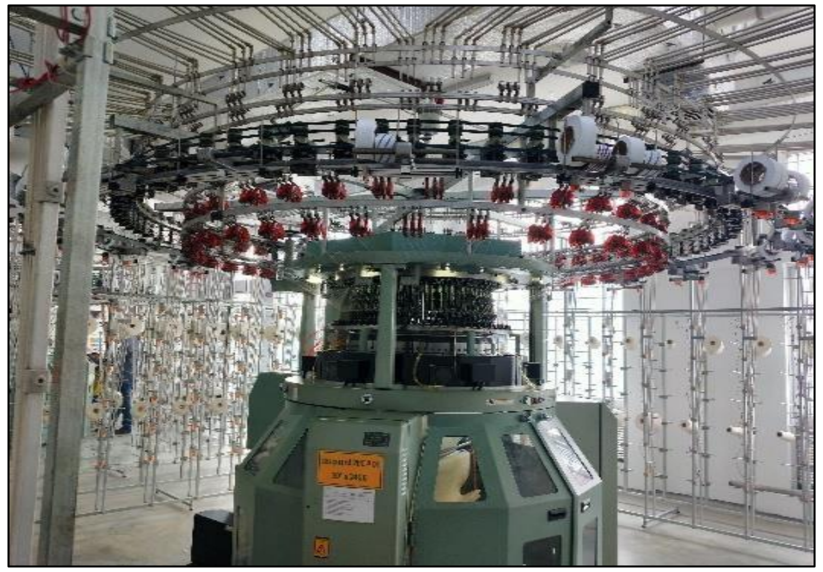
S/J Circular m/c

Fleece Circular m/c

Rib Circular m/c

Interlock Circular m/c

Auto-stripe Circular m/c



S/J Jacquard Circular m/c

D/J Jacquard Circular m/c

Flat Bed Knitting m/c

FEATURES OF CIRCULAR KNITTING MACHINE

1. Circular knitting machine has normally rotating (clockwise) cylindrical needle bed.

2. On circular knitting machines mainly latch needles are used.

3. For single jersey machine holding down sinkers are used, one between every needle space. For double jersey machine, dial needle bed replaces the sinker.

4. Normally stationary angular cam systems are used for needle and sinker.

5. Latch needle cylinder and sinker ring or dial revolve through the stationary knitting cam systems.

6. For single jersey machine, sinker ring which is simply and directly attached to the outside top of the needle cylinder.

7. Stationary yarn feeders are situated at regular intervals around the circumference of the rotating cylinder.

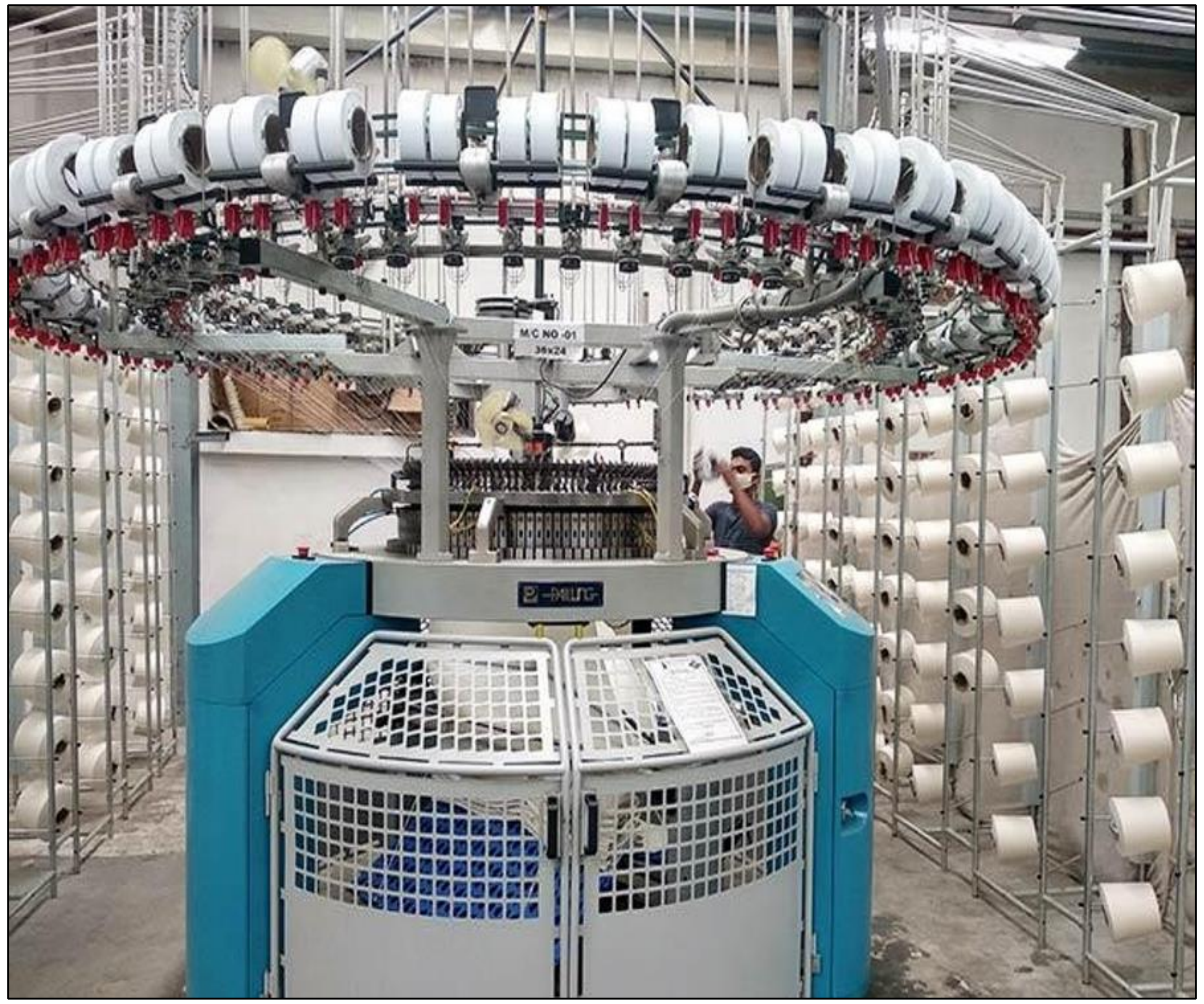
8. Yarn is supplied from cones placed either on an integral over head bobbin stand or on a free-standing creel through tensioners, stop motions and yarn guide eyes down to the yarn feeder guides.

FEATURES OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

1. It is a single-bed circular knitting machine having only one set of needles.
2. The needles are latch-type and mounted vertically in the slots of the needle cylinder.
3. The sinkers are fitted between every two needles and move radially.
4. The machine produces plain fabric (single jersey) which has technical face and back sides.
5. Usually, the diameter of the machine ranges from 12 to 60 inches.
6. The gauge may vary from 14 to 36 needles per inch, depending on yarn count and end use.
7. Cam systems are placed around the cylinder to control needle movement for knit, tuck, and miss stitches.
8. The yarn feeders (generally 90-120 per machine) feed yarns continuously to each knitting head.
9. Sinker ring which is simply and directly attached to the outside top of the needle cylinder.
10. The fabric is drawn down through the center of the cylinder by take-down rollers.
11. The production rate depends on machine diameter, speed (revolutions per minute - RPM), and number of feeders.
12. The machine frame holds the main drive, take-down mechanism, fabric collection, and lubrication systems.
13. Fabrics produced are used for T-shirts, underwear, sportswear, and other lightweight garments.

SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

- Creel
- Cone package holder
- Yarn passing tube
- Tensioner
- Positive feeder
- VDQ pully
- Stop motion device
- Feeder
- Needle
- Cam
- Cam box
- Cylinder
- Sinker
- Fabric spreader
- Take down roller
- Cloth roller
- Lubrication system
- Dust removing fan

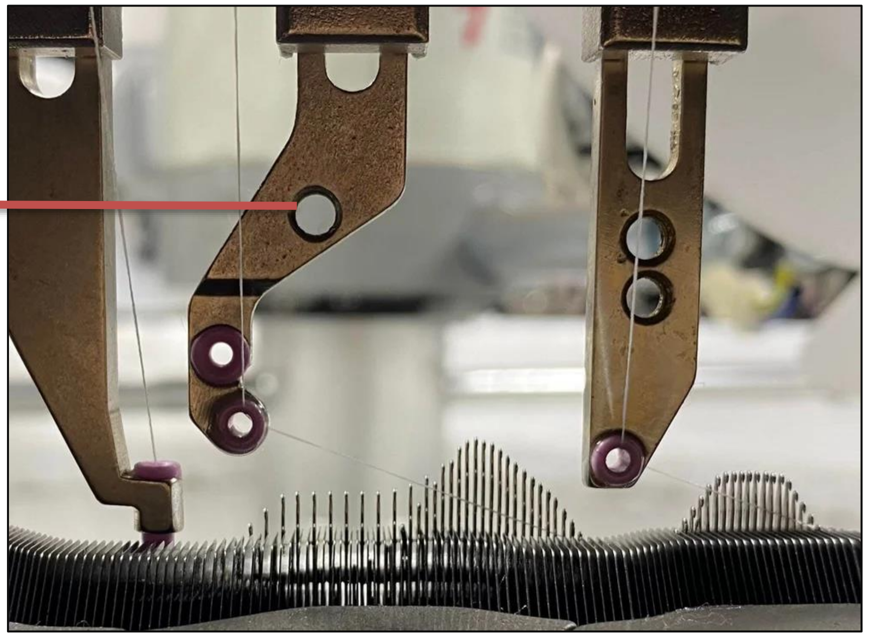


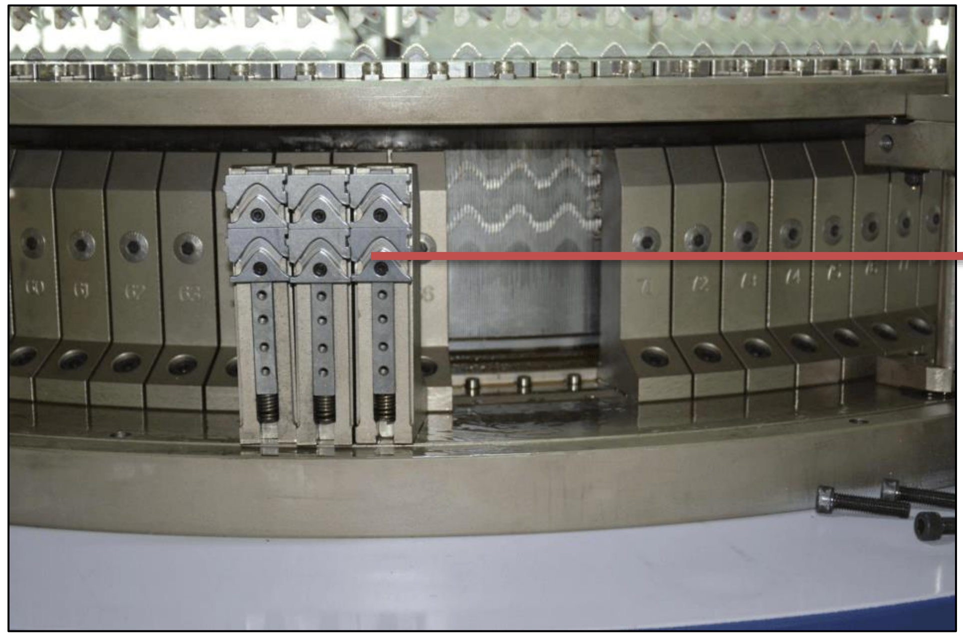
SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

POSITIVE FEEDER

SINGLE JERSEY CIRCULAR KNITTING MACHINE PARTS

YARN FEEDER





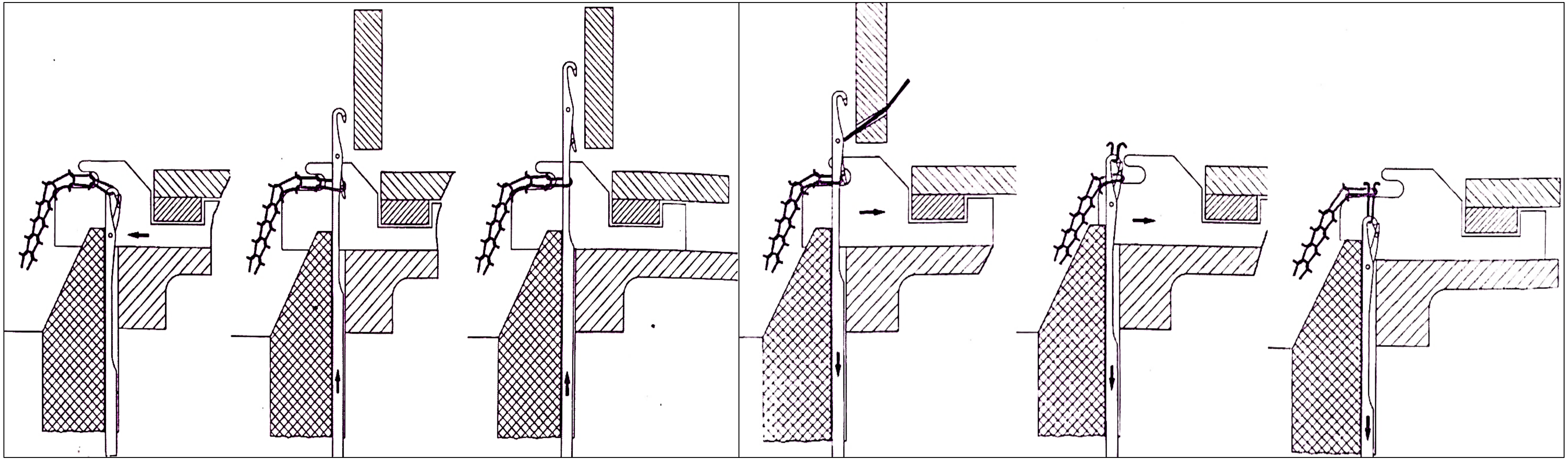
CAM

CYLINDER

SPREADER

https://youtu.be/9M_0l3z7x84?si=BguwdVXdVHNCHv7L

KNITTING ACTION OF SINGLE JERSEY CIRCULAR KNITTING MACHINE



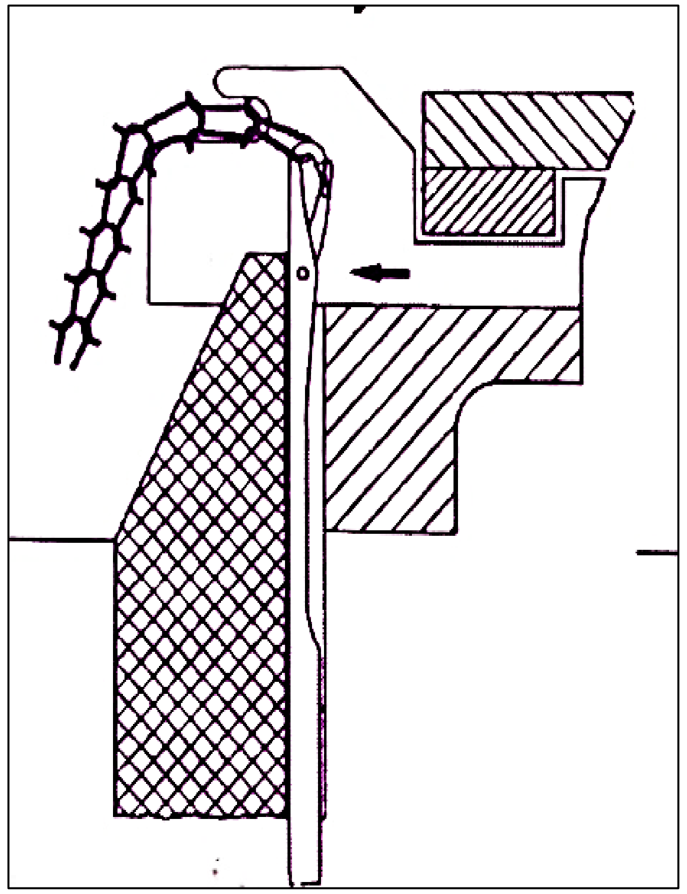
KNITTING ACTION OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

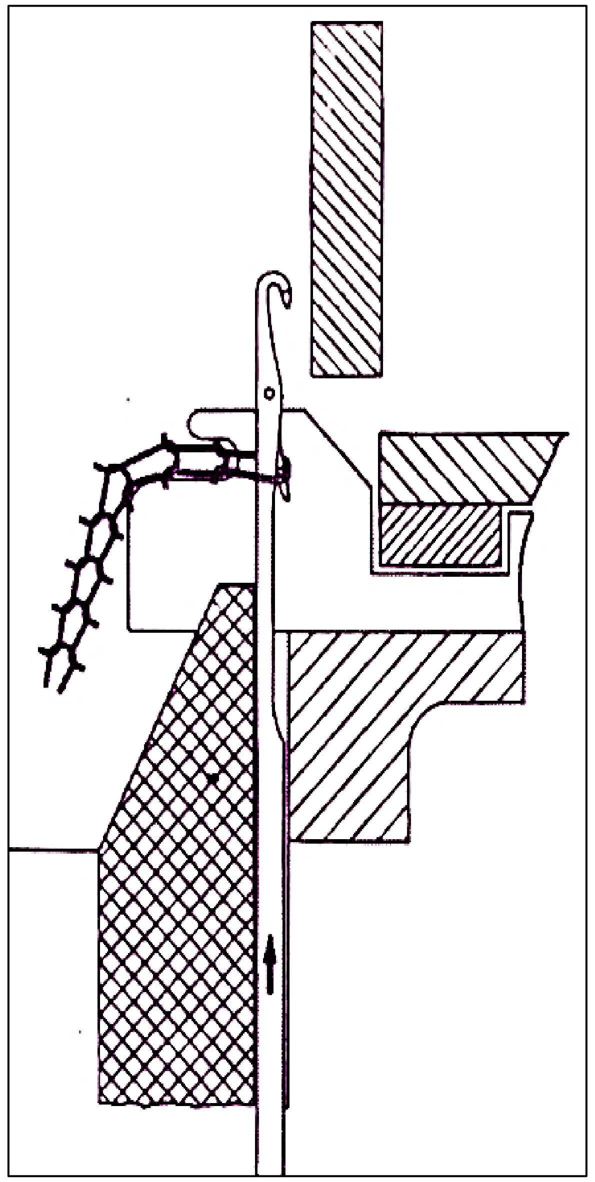
<u>Position 1: Rest-in Position</u>

In the rest position, the *needle is at its starting point.* The top edge of the needle head is *level with the knock-over edge of the sinker.* The *sinker moves towards the center of the cylinder and its throat holds the old loop* and keeps the fabric in position.

<u>Position 2: Tucking-in Position</u>

In this position, the *needle moves upward from the rest position.* During this movement, the *old loop opens the needle latch and rests on the latch.* The yarn feeder keeps the latch open and prevents it from closing. *The sinker does not move during this stage.*





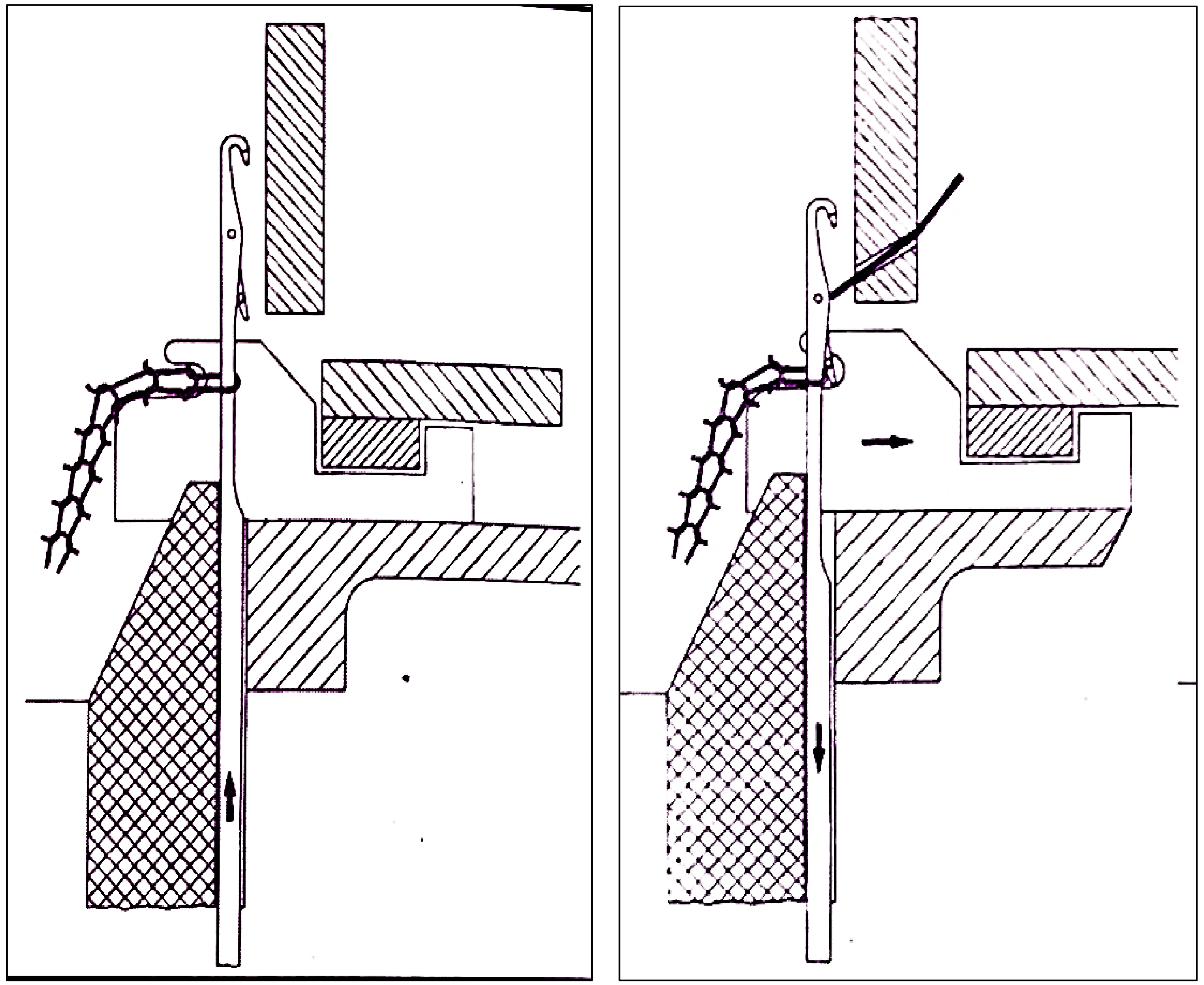
KNITTING ACTION OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

<u>Position 3: Clearing Position</u>

The needle moves further upward and reaches its highest position. *Since the sinker holds the fabric down, the old loop slides along the needle stem and moves below the latch.* The yarn feeder prevents the latch from closing, and the sinker remains still.

<u>Position 4: Yarn Presenting Position</u>

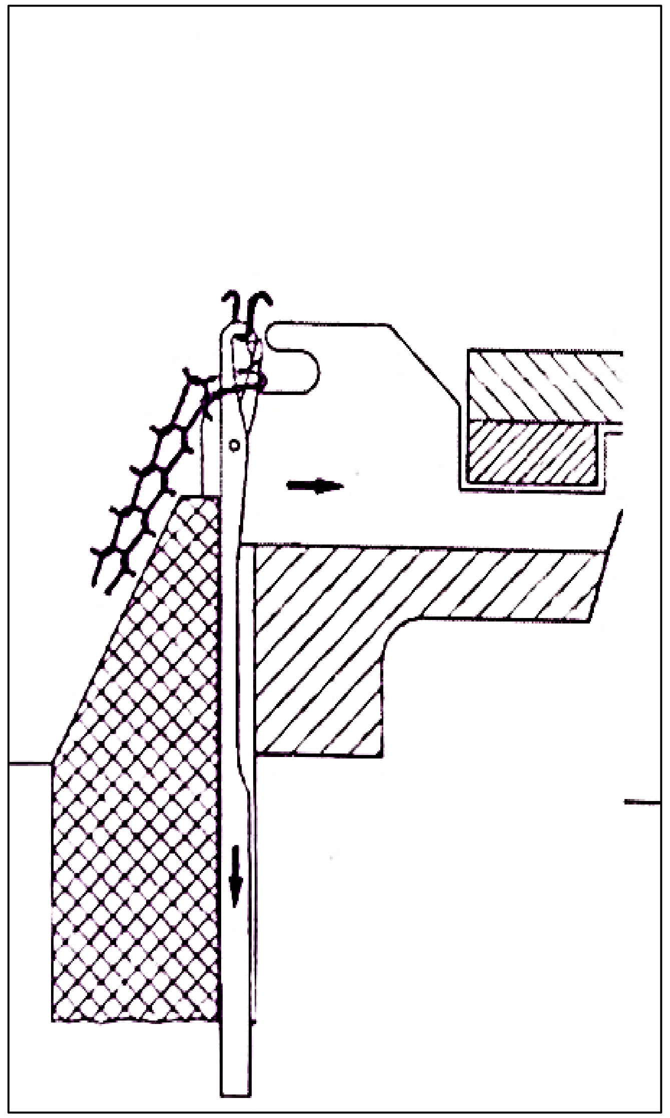
The *needle starts moving downward from the clearing position.* Before the old loop closes the latch, *the yarn feeder places the new yarn inside the needle hook.* After the yarn is fed*, the latch can close during further downward movement.* At this stage, *the sinker moves away from the cylinder center because the fabric no longer needs to be held by the sinker throat.*



KNITTING ACTION OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

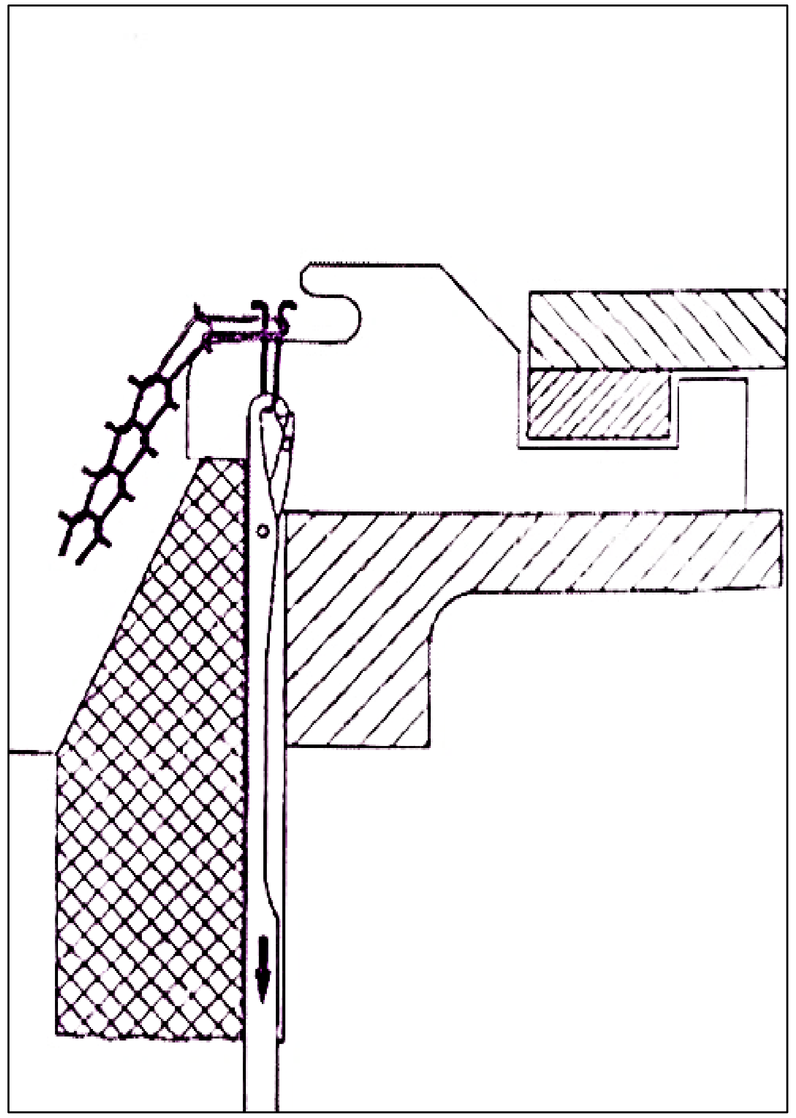
<u>Position 5: Landing Position</u>

The needle continues moving downward from the yarn presenting position. The old loop closes the needle latch, and *the new yarn is trapped inside the closed needle hook.* The sinker reaches its extreme position and *holds the old loop at the knock-over edge so that the new yarn can be pulled through it.*



<u>Position 6: Knocking over Position</u>

The needle moves further downward and pulls the new yarn through the old loop. *The old loop slips off the needle and a new stitch is formed.* The movement of the take-down and guide segments controls the length of the newly formed loop, allowing accurate adjustment of stitch size.

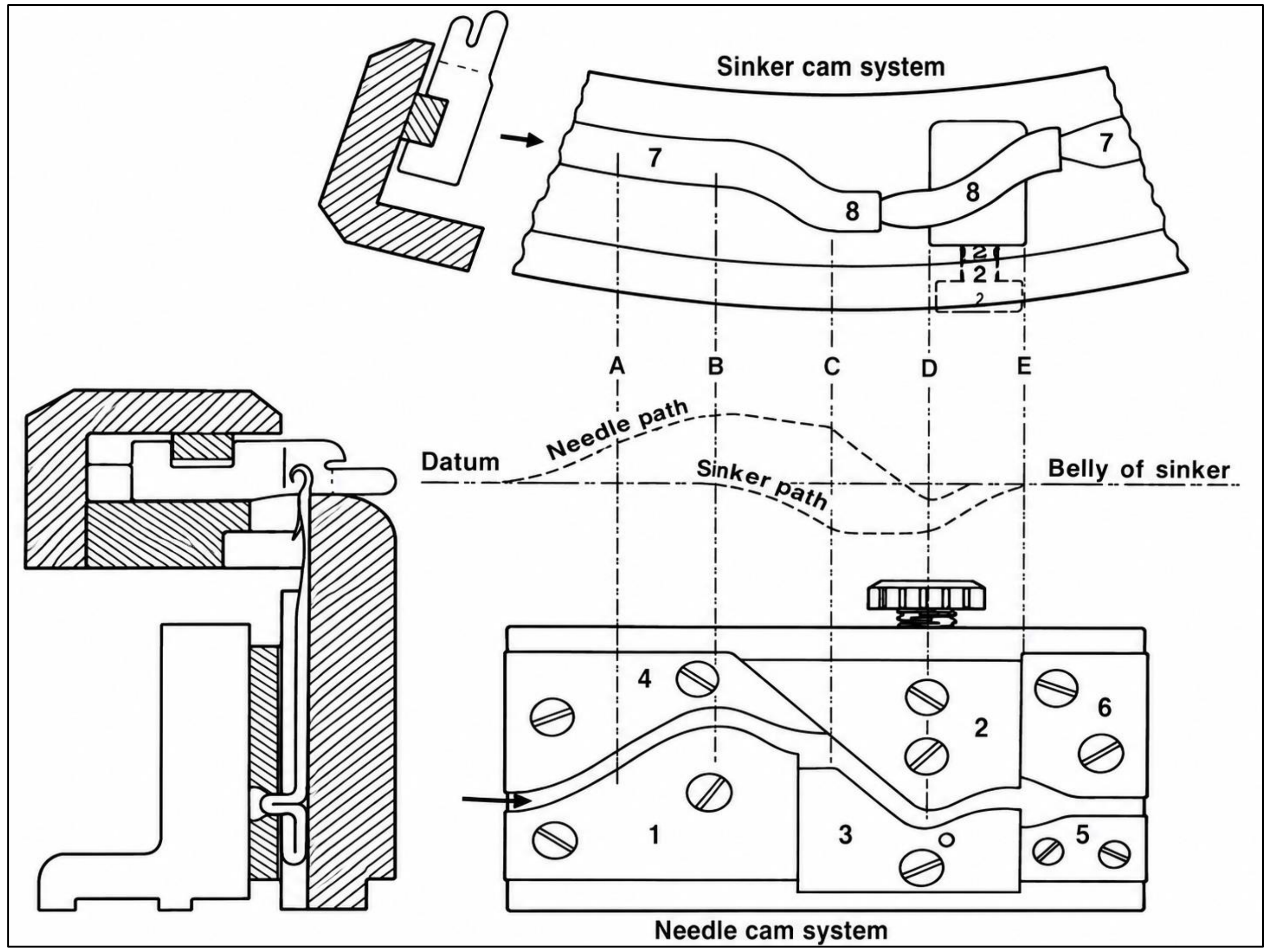


CAM SYSTEM OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

The cam system has two main parts: *Needle cam system & Sinker cam system.*

The needle cam race consists of six main cams:

1. Clearing cam (Raising cam)
2. Stitch cam (Lowering cam)
3. Upthrow cam.
4. Guard cam of clearing cam.
5. Return cam brings the needle back to its position.
6. Guard cam of return cam.



The stitch cam (2) and upthrow cam (3) can be adjusted vertically to change the stitch length.

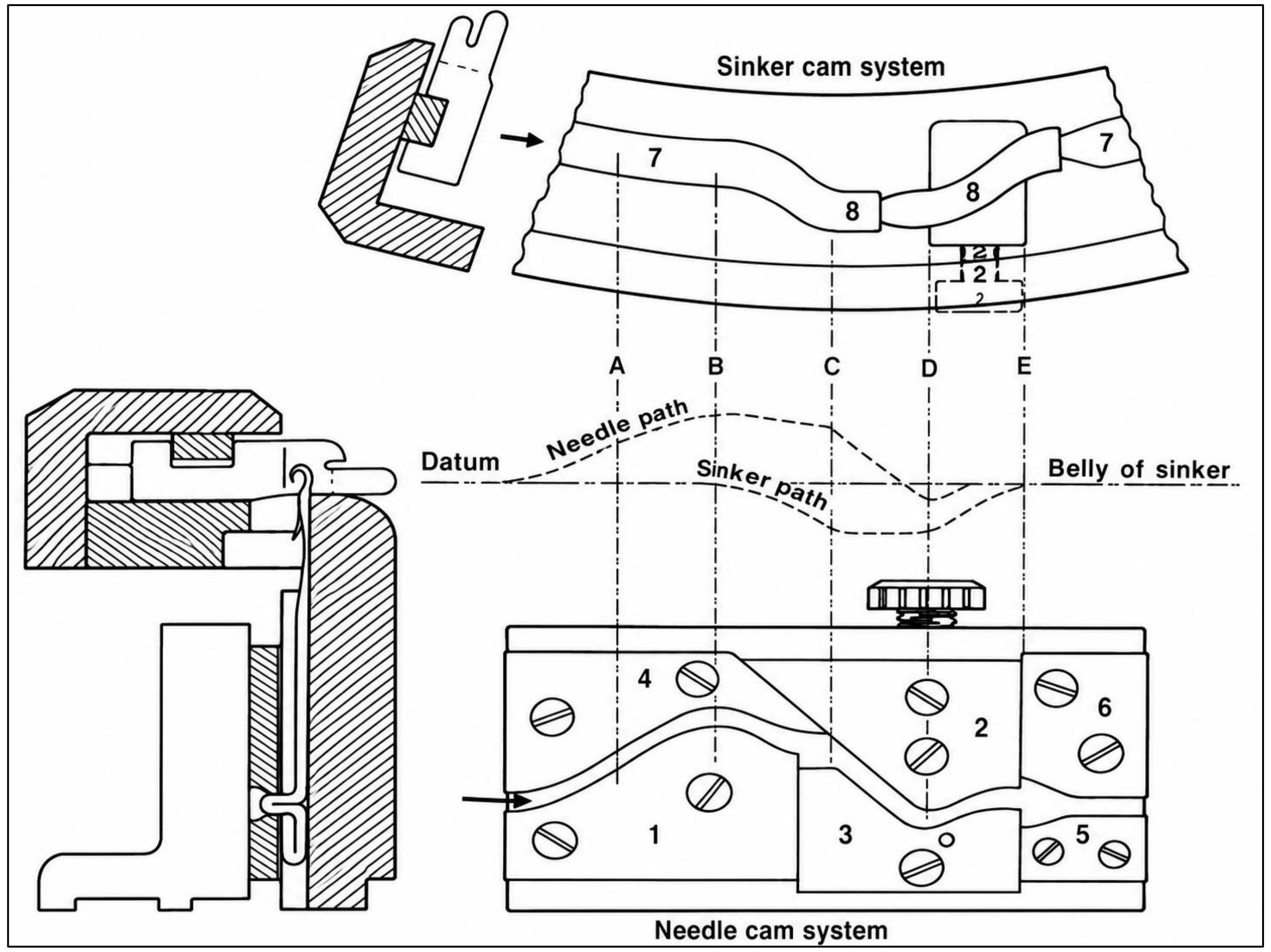
CAM SYSTEM OF SINGLE JERSEY CIRCULAR KNITTING MACHINE

The sinker cam race consists of the followings:

7. The race cam

8. The sinker- withdrawing cam and

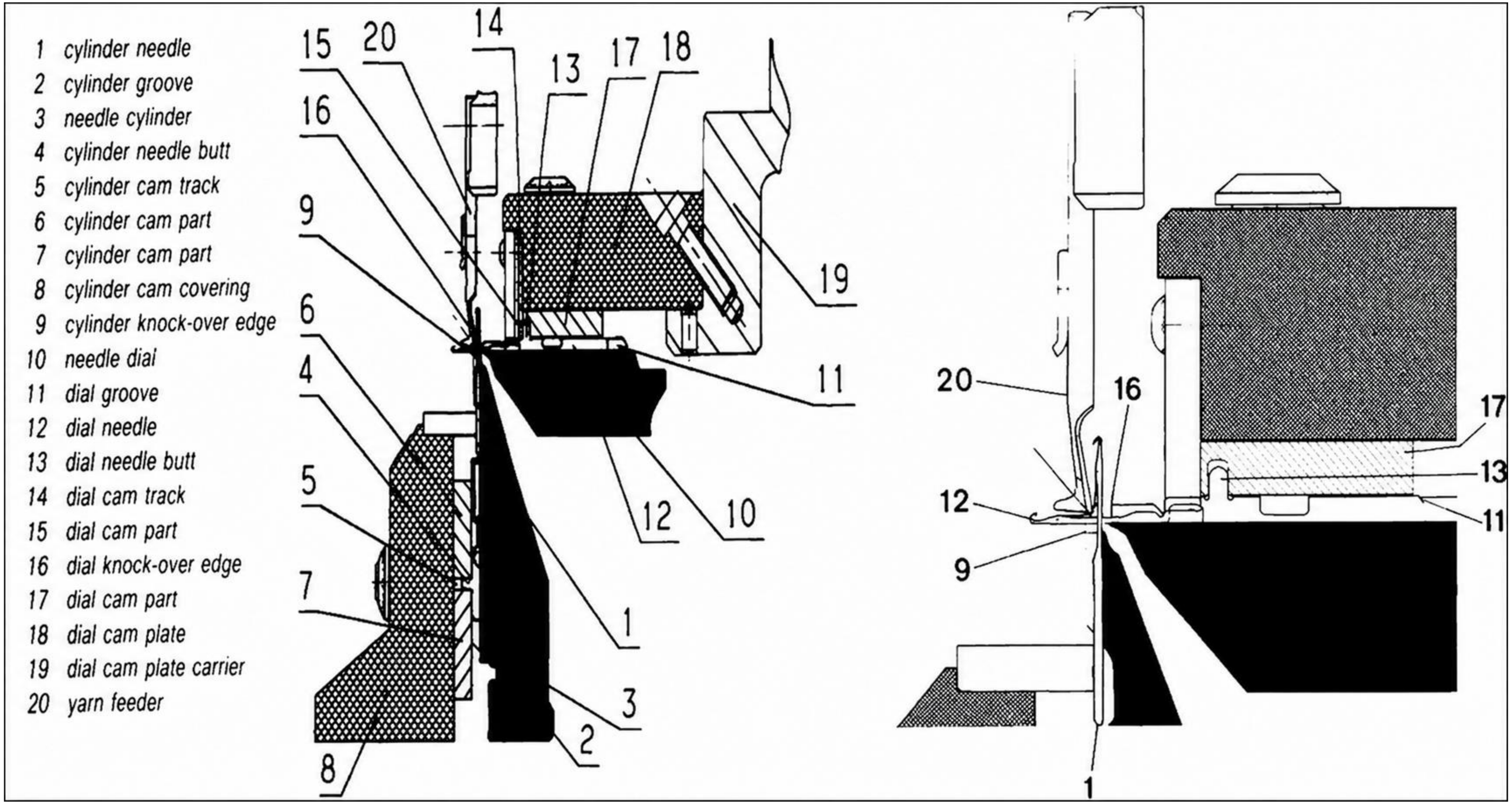
9. The sinker- return cam.



FEATURES OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE

- In this machine group there is one set of needles on the circumference of a vertical cylinder and a second set of needles, arranged perpendicular to the first set and mounted on a horizontal dial.
- On most of the circular knitting machines the cylinder and dial rotate, whereas the cams with yarn feeder guides are stationary.
- The following figure shows a cross-sectional view of the region containing the knitting elements of a rib circular knitting machine. The set-up of the cylinder (3) with its knitting elements (1 to 9) is the same as with plain circular knitting machines. In a horizontal dial (10) grooves (11) are milled in. The latch needles (12) are housed and guided in these grooves. The dial needle (12) obtains its motion for stitch formation through its butt 13, which extends into a cam track (14). This cam track (14) is formed by the cam parts (15) and (17), which in turn are fixed to a dial cam plate (18). During the rotation of the cylinder and the dial the cylinder needle (1) is moved vertically and the dial needle 12 is moved horizontally, corresponding to the shape of the cam track in the cylinder and dial cams.
- In a gauge range from 5 to 20 needle per inch.

FEATURES OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE



NEEDLE TIMING

Needle timing is the *position difference between the dial needle knock-over point and the cylinder needle knock-over point* in a circular knitting machine.

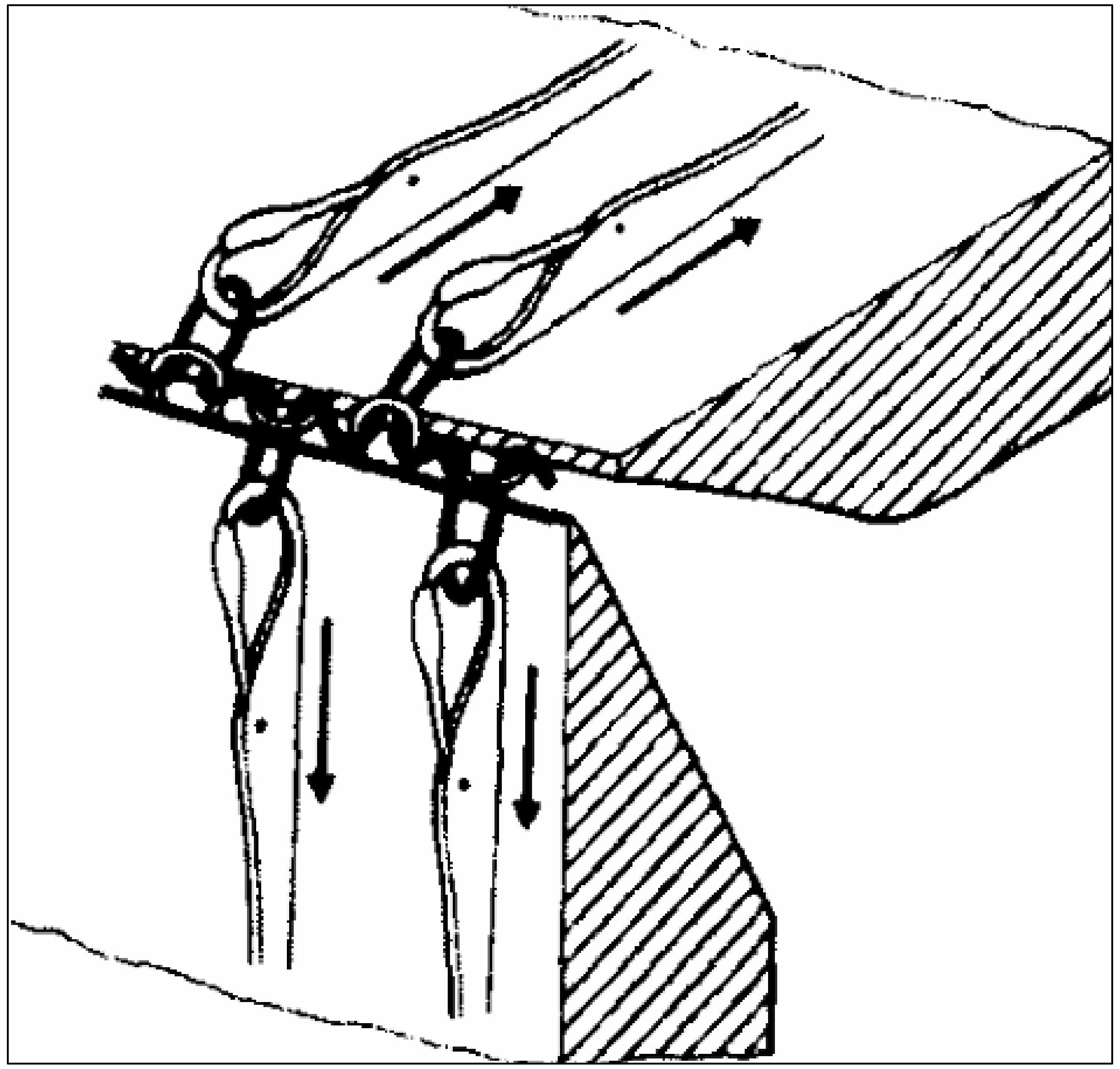
It is adjusted by changing the *position of the dial cam plate compared with the cylinder cam.* Individual adjustment can be made by changing or replacing the stitch cam profile.

There are two types of needle timing:

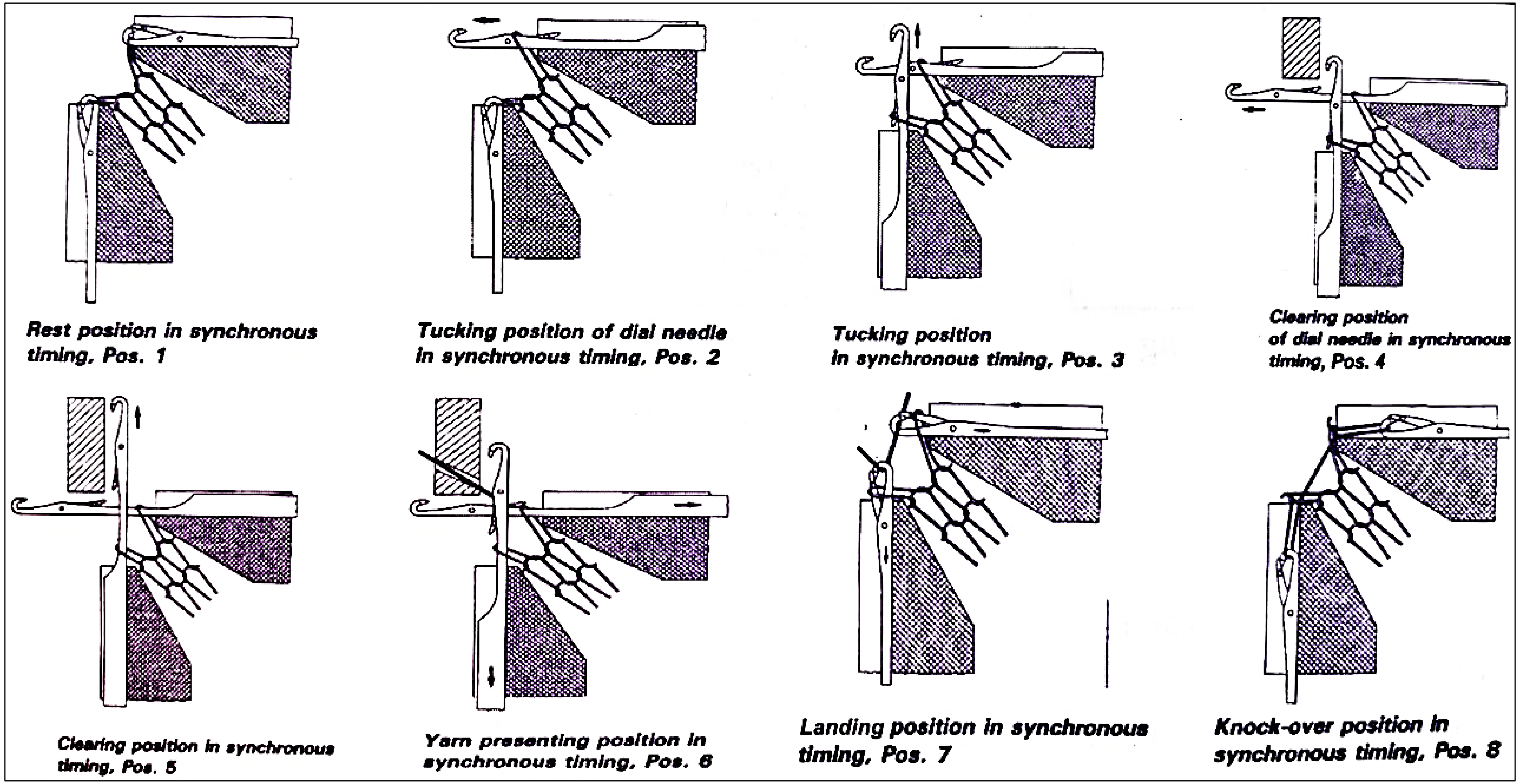
- ❑ Synchronized timing: *Dial and cylinder needles knock over at the same time.*
- ❑ Delayed timing: *Dial needle knock-over occurs later than the cylinder needle.*

SYNCHRONIZED TIMING

- In synchronized timing, the *cylinder needles and dial needles knock-over their loops at the same time.*
- Both needles reach the knock-over point together, and *their knock-over depths are equal.*
- During loop formation, the *yarn is pulled in two opposite directions by the cylinder and dial needles, which creates high tension in the yarn.*
- This timing can be used for *most rib and interlock structures.*
- Fabrics produced with synchronized timing are *usually looser and have uneven stitches* because of the high yarn tension.

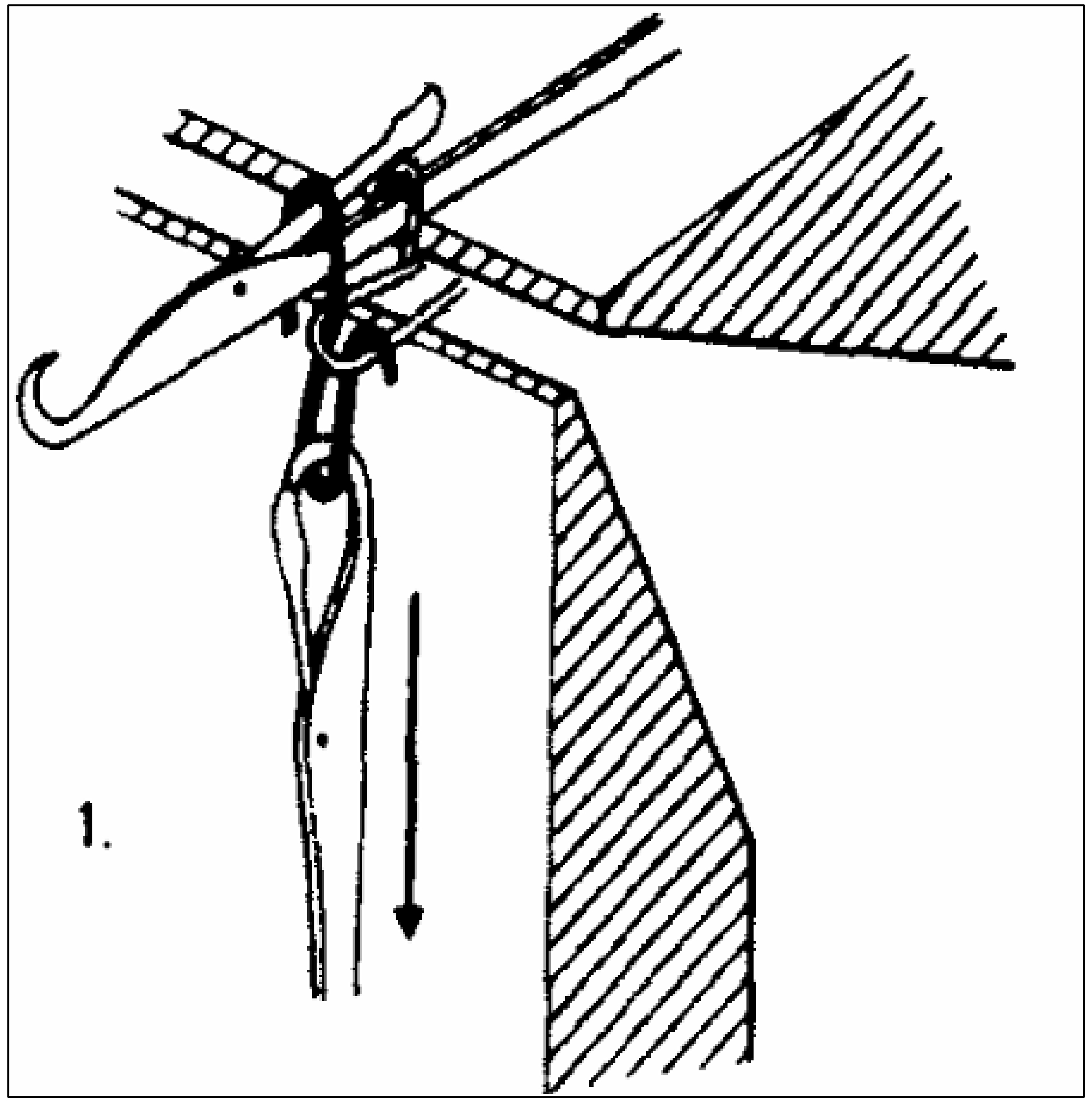


KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN SYNCHRONOUS TIMING



DELAYED TIMING

- In delayed timing, the *dial needles knock-over their loops after the cylinder needles.*
- The *cylinder needles first pull the loops* and *then rise slightly to reduce yarn tension.* After this, the *dial needles complete their knock-over process.*
- Because of this delay, the *dial loops become longer than the cylinder loops.*
- The fabric produced with delayed timing is:
  - ✓ Tighter in structure, even in stitch formation
  - ✓ More rigid and stable, heavier and wider
  - ✓ Produced with less strain on the yarn



DELAYED TIMING

Why are dial loops longer?

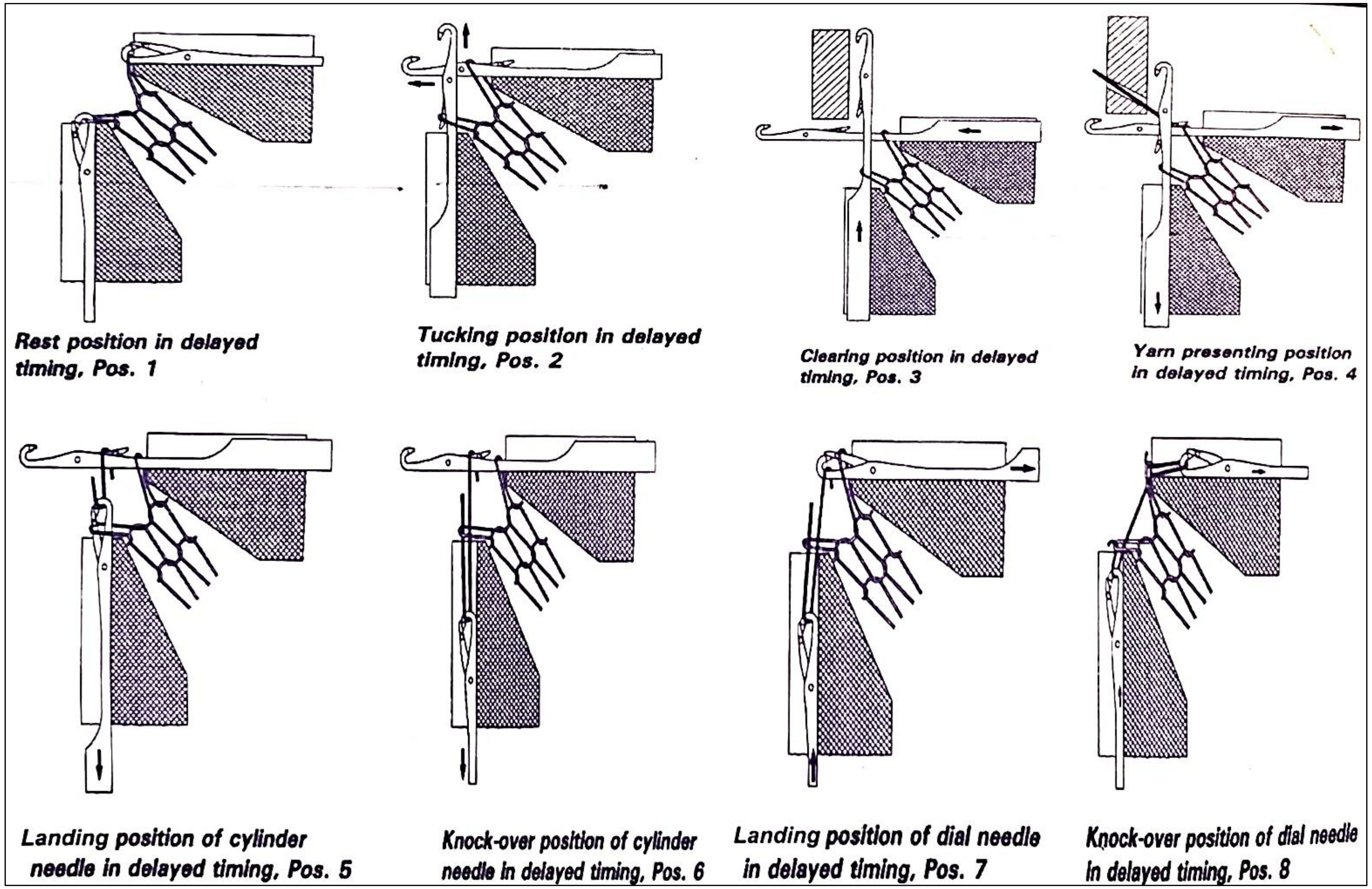
Because the dial needles have extra yarn available. During delayed timing: The cylinder needles have already drawn some yarn. The dial needles then pull their loops and receive a small amount of extra yarn ("robbed" from cylinder loops). Therefore, the dial loop length becomes larger.

Then how are stitches more-even?

Normally, in synchronized timing: Both needle sets pull the yarn at the same time. The yarn tension is very high and unequal forces can create irregular loops.

In delayed timing: Cylinder and dial needles work separately instead of pulling against each other at the same moment. The yarn tension is reduced. The loops form more smoothly and consistently. So, even stitches do not mean cylinder and dial loops are the same size. It means the stitches are more regular and stable.

KNITTING ACTION OF DOUBLE JERSEY CIRCULAR KNITTING MACHINE IN DELAYED TIMING



FLAT BED KNITTING MACHINE

1. The machine consists *of two straight needle beds* arranged *horizontally and facing each other*, forming a *V-shape* (in a V-bed machine).
2. It is used mainly for producing *weft-knitted fabrics* such as *rib, interlock, and purl* structures.
3. Each bed contains *latch needles* which can be individually controlled for stitch selection.
4. *Cams* mounted on the *carriage* move across the needle beds to operate the needles and form loops.
5. The *carriage* carries the cam system and moves *to and fro (bi-directional cam system)* across the beds to complete each course.
6. Yarn is fed to the needles through a *yarn carrier* mounted on the carriage.
7. The *machine width* is specified in terms of the *working width of the needle beds*, typically *36 - 100* inches.
8. *Gauge* (number of needles per inch) may range from *3 to 18 G*, depending on fabric type and yarn count.
9. The machine can be *hand-driven or power-driven*, depending on model and production requirements.
10. *Fabric take-down* is achieved by *rollers or weights pulling the fabric downward.*
11. Flat bed machines are well-suited for *fashion garments, sweaters, collars, ribs, and fully-fashioned* components.

FLAT BED KNITTING MACHINE

12. The structure of the machine allows *easy patterning* and *individual needle selection* by electronic or mechanical means.

13. Stitch cams determine the depth of loop formation, and *tuck or miss cams* can be used for pattern variation.

13. The machine offers *high fabric dimensional stability* and good control over fabric width and shape.

14. The flat machine is the most versatile of weft knitting machines, its stitch potential includes *needle selection on one or both beds, needle-out designs, striping, tubular knitting, changes of knitting width and loop transfer.*

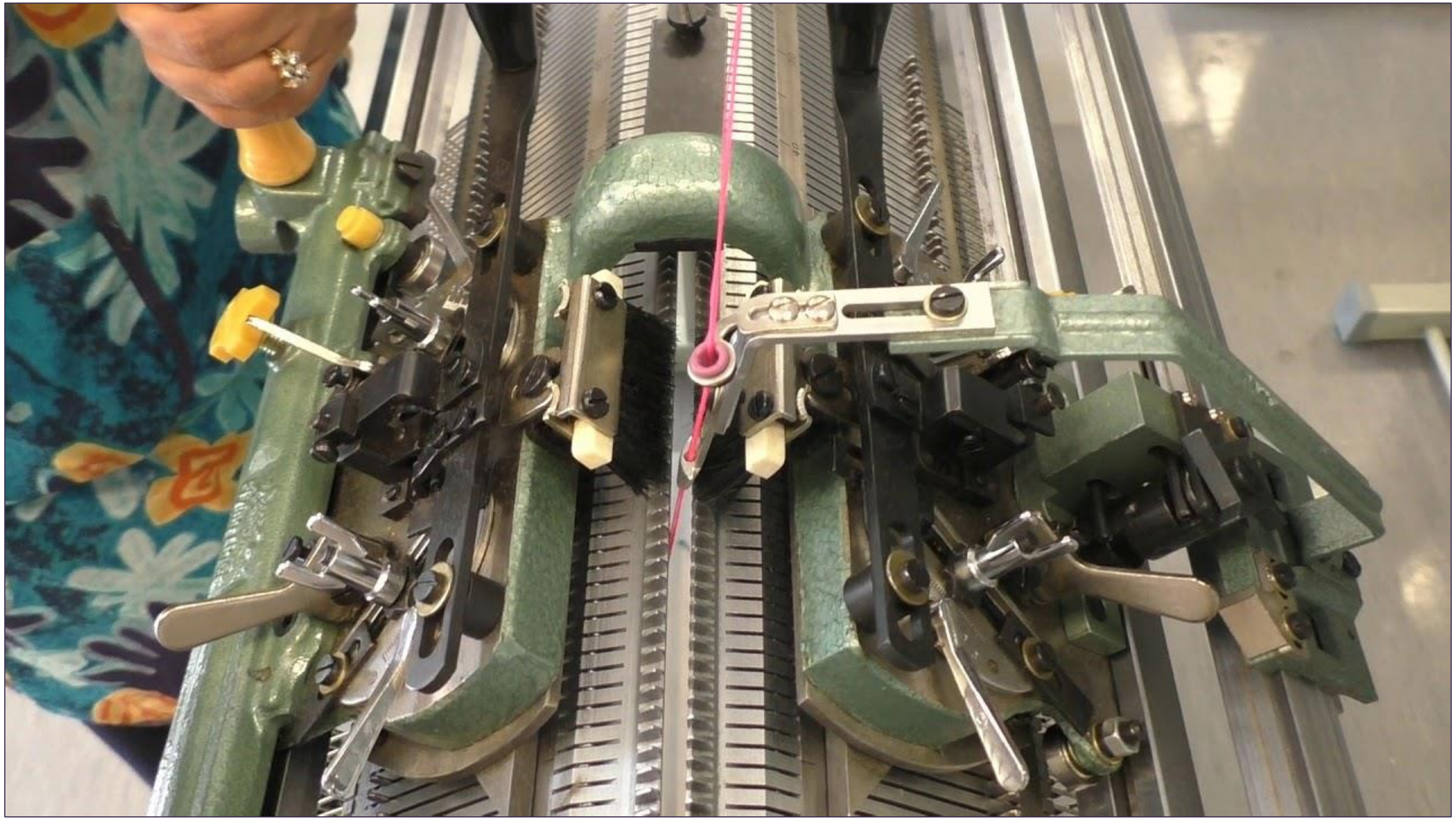
15. A wide range of yarn counts may be knitted per machine gauge including a number of ends of yarn in one knitting system, *the stitch length range is wide and there is the possibility of changing the machine gauge*.

16. The operation and supervision of the machines of the simpler type is relatively less arduous than for circular weft knitting machines.

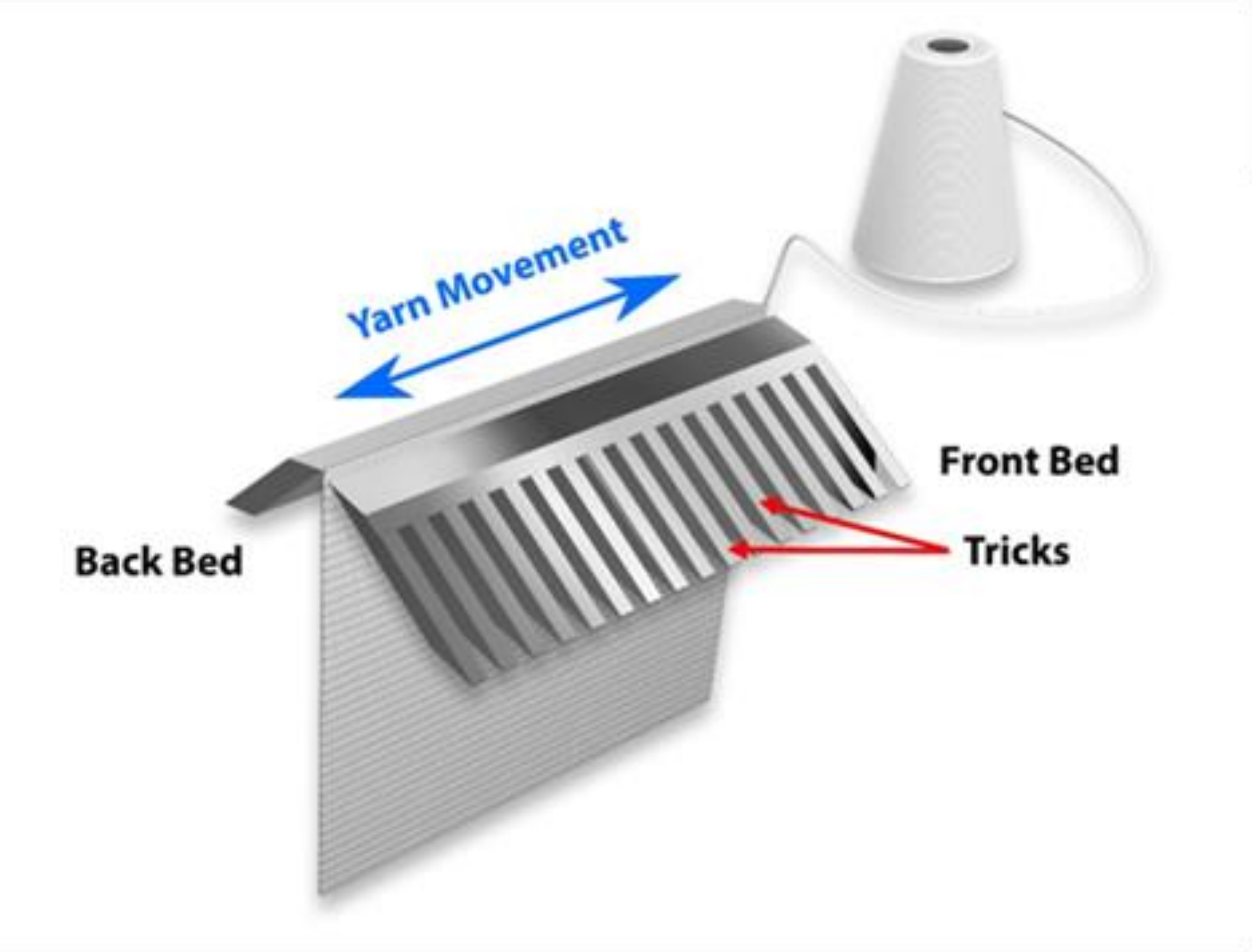
17. The *number of garments or panels simultaneously knitted across* the machine is dependent upon its knitting width, yarn carrier arrangement, yarn path and package accommodation.

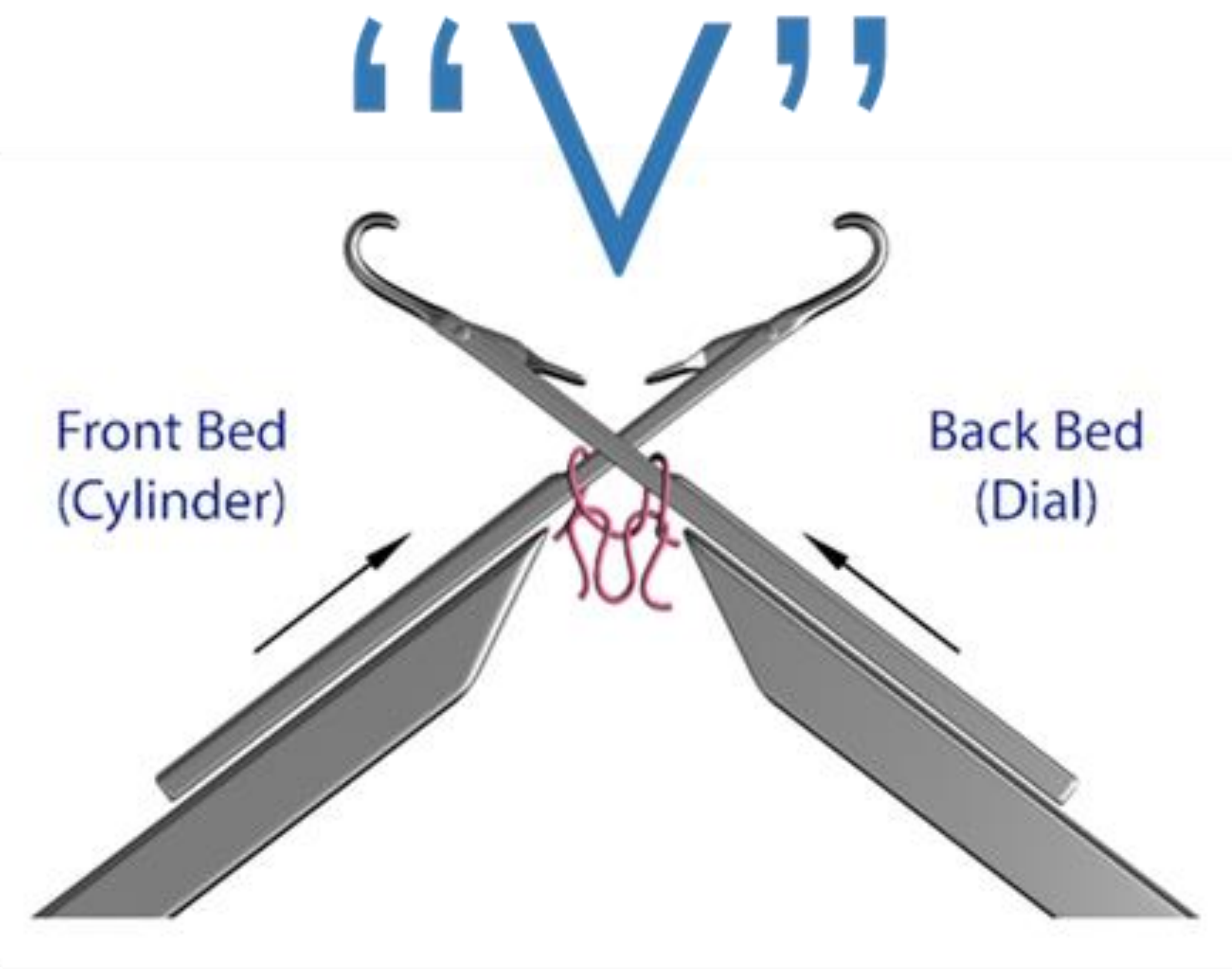
18. Versatile knit structures can be produced *by changing the selection of cam carriage.*

V BED KNITTING MACHINE



V BED KNITTING MACHINE



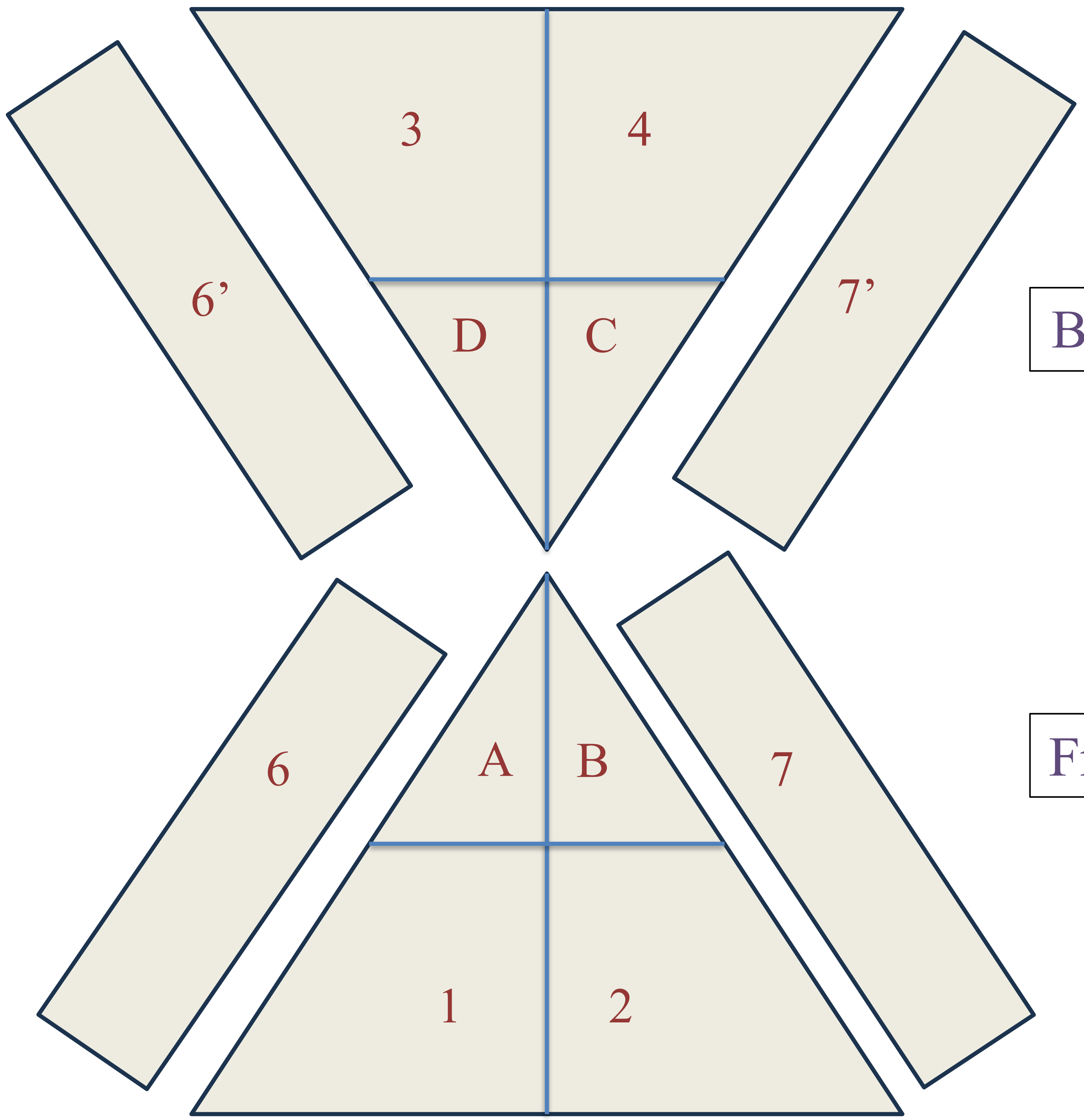


V BED KNITTING MACHINE

1,2,3,4 - Raising Cam

A, B, C, D - Tuck cam/Cardigan cam

6,6’,7,7’- Stich cam

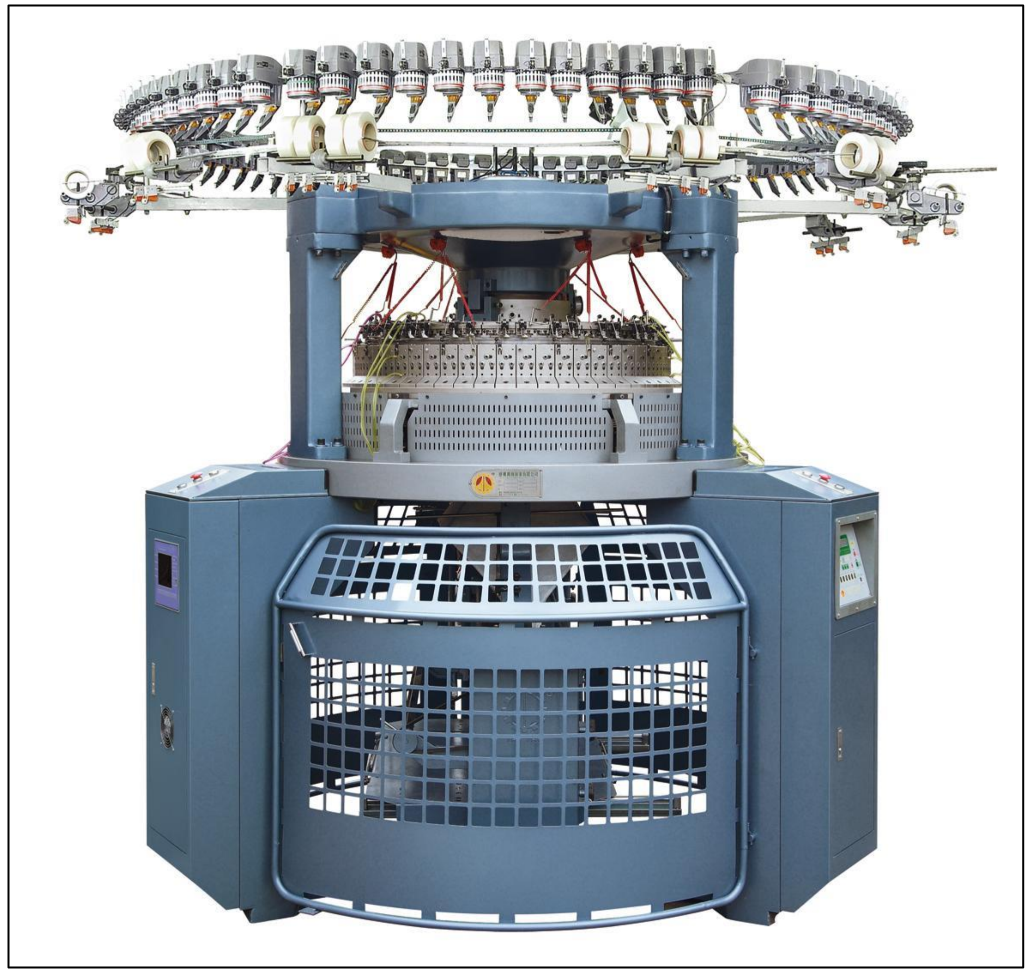


Back needle bed

Front needle bed

TYPES OF WEFT KNITTING MACHINE BASED ON END PRODUCT

- Fabric length machine
- Garment length machine



FEATURES OF FABRIC LENGTH MACHINE

- *Large diameter, circular, latch needle machine* knit fabric at high speed.
- The fabric is *manually cut away from the machine*, usually *in roll form*, after a convenient length has been knitted.
- Most fabric is knitted on circular machines, either single cylinder (Single jersey) or "Dial and cylinder" (Double jersey).
- *Unless used in tubular body* width, the fabric requires *splitting into open width.*
- *Fabric is finished on continuous finishing equipment and is cut-and-sewn into garments.*
- Generally, cam settings and needle set outs are not altered during the knitting of the fabric.
- The productivity, versatility and patterning facilities of fabric machines vary considerably.

FEATURES OF GARMENTS LENGTH MACHINE

- It includes straight bar frames, and Circular garment machine.
- These machines are *coarser gauge machines (Low gauge)* than fabric machines.
- They produce both outerwear, and innerwear.
- Garments may be knitted in *either tubular or open width form*; in the later case, more than one garment panel may be knitted simultaneously across the knitting bed.
- The fabric take down mechanism is adaptable with variable rates of production during the knitting sequence, and on some machines be able to assist both in the setting up on empty needles and take away of separate garments or pieces on completion of the sequence.
- The fabric take down mechanism is more sophisticated than for continuous fabric knitting.

NEEDLE GAITING

The way of two beds of knitting needles and their housings are *aligned with each other in a dial and cylinder* of circular knitting machine or flat bed knitting machine.

Generally, two types of needle gaiting are used. These are:-

- Rib gaiting
- Interlock gaiting

NEEDLE GAITING

1. Rib Gaiting: When the needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed at alternate position with each other is known as Rib gaiting.

Fig: Rib Gaiting

NEEDLE GAITING

2. <u>Interlock Gaiting</u>: When needles in the cylinder and dial in circular knitting machine or front needle bed and back needle bed in flat knitting machine are placed at face-to-face position with each other is known as Interlock gaiting.

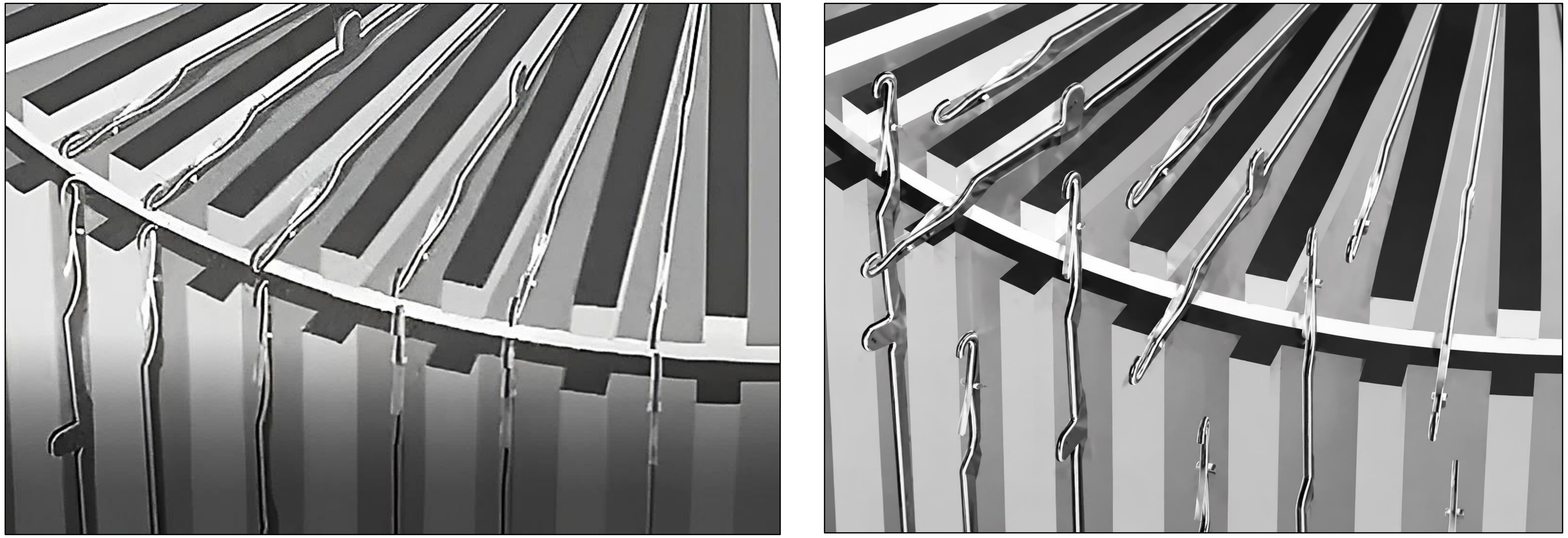


Fig: Interlock Gaiting

INTERLOCK CIRCULAR KNITTING MACHINE

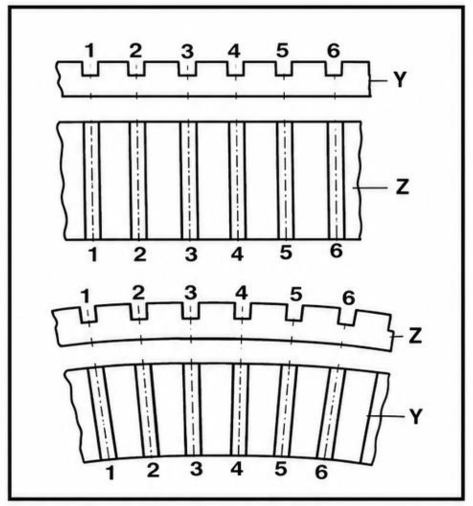
Interlock fabric is mainly produced on special cylinder and dial circular machines and some double-system Vee-bed flat machines.

Requirements of an Interlock Machine:

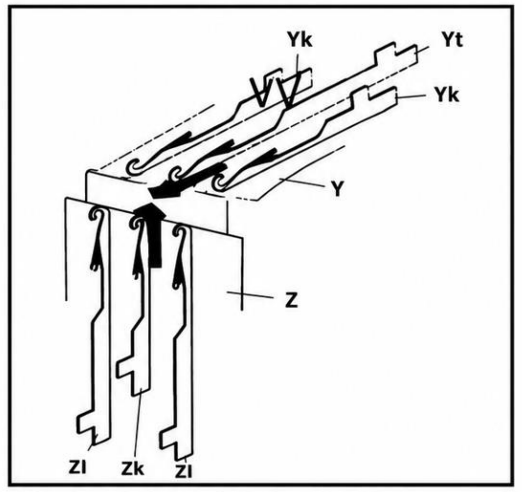
The *needles in the cylinder and dial must be exactly opposite each other.* This is called interlock gating. At each feeder, only one set of needles works, while the other set follows a non-knitting path. Each needle bed has two separate cam systems: One controls half of the needles at one feeder. The other controls the remaining needles at the next feeder. Needles are arranged in an alternate pattern, so opposite needles do not knit together at the same feeder. Conventional Interlock Machine uses two types of needles: Long needles & Short needles. Long needles work at the first feeder. Short needles work at the second feeder. Needles are arranged alternately, with a long needle opposite a short needle.

INTERLOCK CIRCULAR KNITTING MACHINE

At one feeder, long cylinder and dial needles form stitches. At the next feeder, short cylinder and dial needles form stitches. The needles not knitting at a feeder move through a non-knitting track. Modern machines usually use needles of the same length.



Interlock gaiting for cylinder and dial on a rib circular knitting machine in front view (above) and top view (below)



Interlock position of dial and cylinder needles

CAM SYSTEM OF INTERLOCK KNITTING MACHINE

Cylinder cam system:

The cylinder needle cam system consists of the followings:

- ❖ A, is a clearing cam which lifts the needle to clear the old loops.
- ❖ B and C, are the stitch and guard cams respectively and are vertically adjustable for varying stitch length.
- ❖ D, is upthrow cam, to raise cylinder needle whilst dial needle knocks over.
- ❖ E and F, are the guard cams, to complete the track.
- ❖ G and H, provide the track for the idling needles.

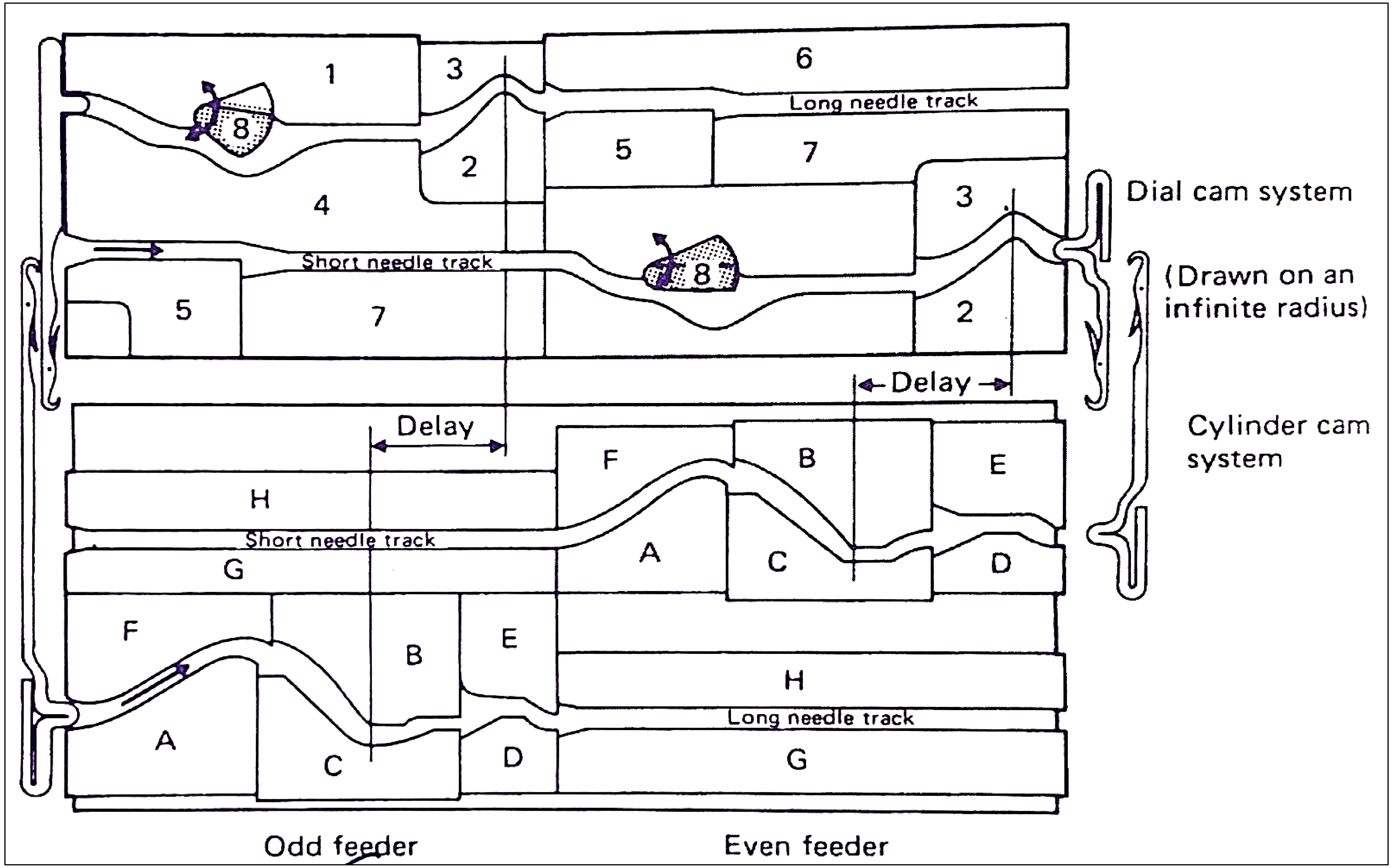
CAM SYSTEM OF INTERLOCK KNITTING MACHINE

Dial cam system:

The dial needle cam system consists of the followings:

- 1, is a raising cam to the tuck position only.
- 2 and 3, are the adjustable dial knock-over cams.
- 4, is a guard cam to complete the track. (Drawn on an infinite radius) Cylinder cam system
- 5, is an auxiliary knock-over cam to prevent the dial needle re-entering old loop.
- 6 and 7, provide the track for the idling needles.
- 8, is a swing type clearing cam, which may occupy the knitting position as shown at feeder 1 or the tuck position as shown at feeder 2.

CAM SYSTEM OF INTERLOCK KNITTING MACHINE



PURL KNITTING MACHINE

In a flat links-links machine, double-headed latch needles are used. These needles can move between two needle beds because the slots of both beds are placed opposite each other.

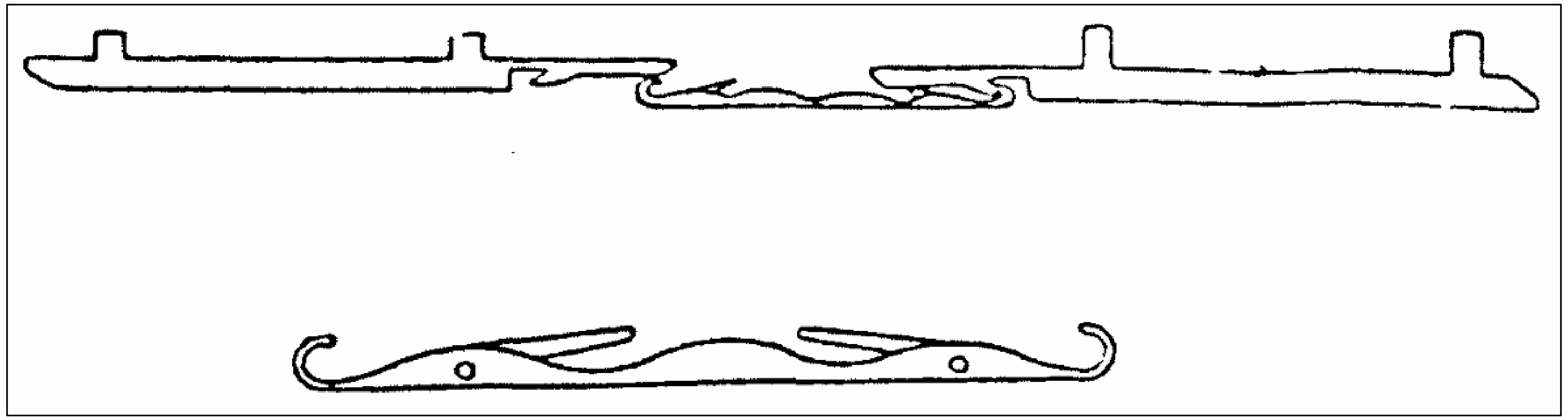
The needle beds have special knock-over teeth that hold the old loops during stitch formation.

Each needle has two sliders:

Slider M controls the left head of the needle.

Slider N controls the right head of the needle.

These sliders move the needle back and forth to form stitches.



PURL KNITTING MACHINE

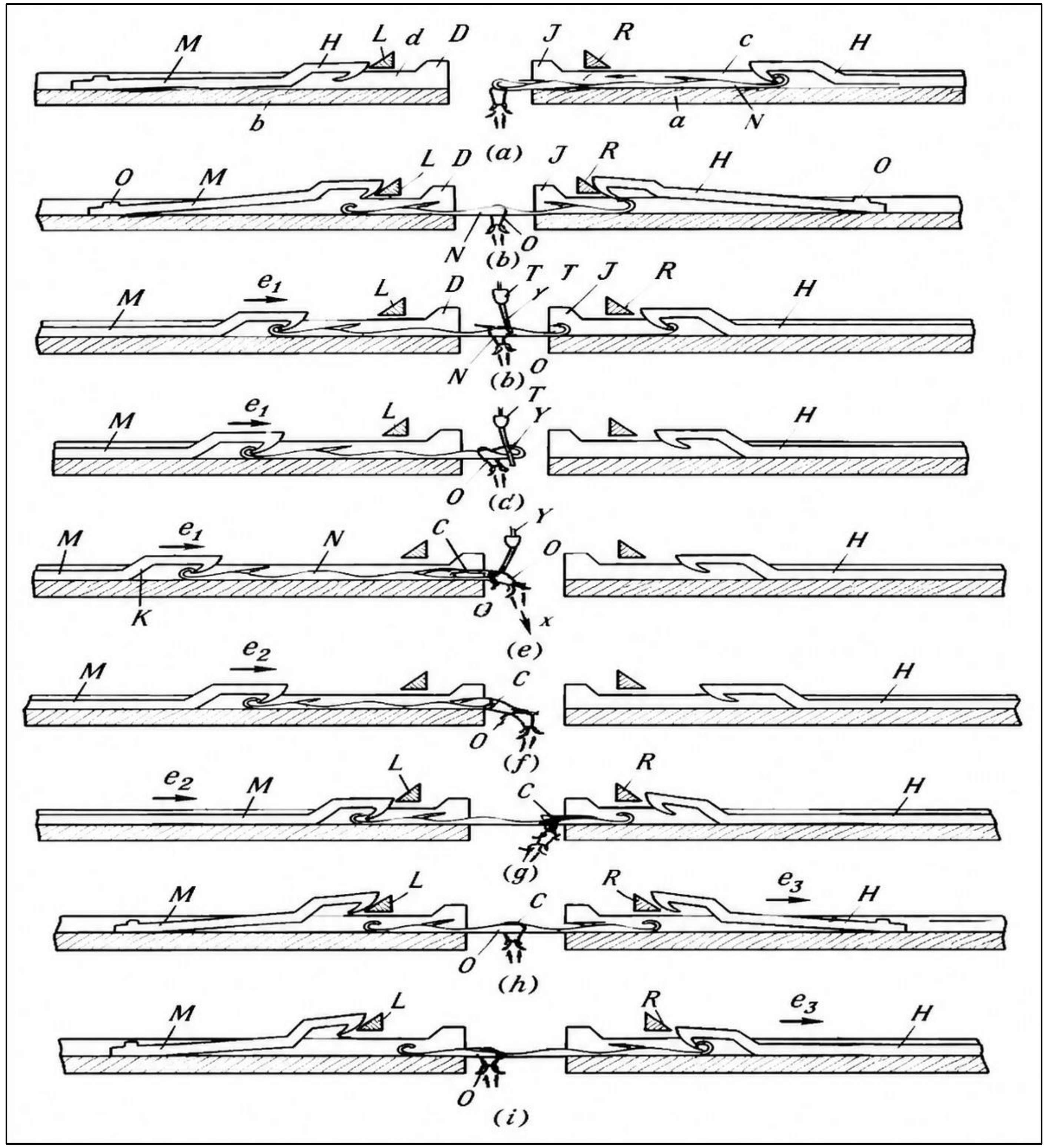
Stitch Formation Process

1. Needle Movement and Loop Transfer
   - A slider moves the needle from one bed to the other.
   - The old loop opens the needle latch and moves to the middle of the needle.
2. Yarn Laying
   - The yarn is placed on the needle hook.
   - The yarn must pass under the hook correctly and should not be caught by the latch.
3. Underlapping and Pressing
   - The needle moves forward.
   - The old loop is held by the knock-over teeth.
   - The old loop closes the latch, trapping the new yarn inside the needle hook.

PURL KNITTING MACHINE

4. Landing, Joining, and Casting-off
    - The needle continues moving.
    - The old loop moves over the latch and joins with the new yarn.
    - The old loop is removed, and a new loop is formed.
5. Formation of Purl Stitch
    - When the left needle head forms the loop, the old loop is cast off to the left side.
    - When the right needle head forms the loop, the old loop is cast off to the right side.
    - This opposite loop formation creates a purl stitch.

PURL KNITTING MACHINE



THANK YOU !